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Computer-Assisted Theorem Proving: 7CCSACTP

Exercise Sheet 7

Important Note: Please bring a laptop with Isabelle/HOL installed to the Large Group Tutorial (LGT). Performing these proofs in Isabelle will be an integral part of the tutorial.

Exercise 7.1 Abstract Data Types via Locales: Map ADT

In the lecture, we introduced Isabelle *locales* to formalise an Abstract Data Type (ADT) for sets. In this exercise, you will define a parameterised locale for a *Map ADT* that maps keys of type $'k$ to values of type $'v$.

The locale should declare: 1. An empty map constant *empty_map*, 2. An update operation *update* :: $'k \Rightarrow 'v \Rightarrow 'm \Rightarrow 'm$, and 3. A lookup operation *lookup* :: $'k \Rightarrow 'm \Rightarrow 'v$ *option*.

It must enforce the standard lookup-after-update invariants as locale assumptions.

```
locale Map_ADT =
  fixes empty_map :: 'm
    and update :: 'k ⇒ 'v ⇒ 'm ⇒ 'm
    and lookup :: 'k ⇒ 'm ⇒ 'v option
  assumes lookup_empty: lookup k empty_map = None
      and lookup_update_same: lookup k (update k v m) = Some v
      and lookup_update_diff: k ≠ k' ⟹ lookup k (update k' v m) = lookup k m
```

Prove that inserting a key-value pair and immediately overriding the same key yields the same lookup result.

```
lemma (in Map_ADT) lookup_override:
  lookup k (update k v2 (update k v1 m)) = Some v2
  sorry
```